



TIMELY Bench for Temporal Clinical Reasoning in Critical Care

**MSc Applied Statistical Modelling and Health Informatics
Research Project
King's College London**

Student: Haoyu Wang

Supervisor:

Dr. Zina Ibrahim

Dr. Linglong Qian

haoyu.7.wang@kcl.ac.uk

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Abstract

Background: Temporal clinical reasoning requires interpreting partially observed patient trajectories, but most clinical AI benchmarks test static prediction, medical question answering, or workflow execution. **Methods:** Using MIMIC-IV ICU stays, we developed TIMELY Bench, a 168-hour anchor-bounded benchmark for AKI, delirium, sepsis, and stroke. The frozen sample included 12,000 instances from 9,587 stays, expanded to 53,070 `full_multimodal` prompts per LLM provider; structured baselines were evaluated separately on Branch A summaries and Branch B1 hourly sequences. We evaluated temporal localization, threshold reasoning, trend interpretation, diagnostic discrimination, contrastive reasoning, and evidence attribution. **Results:** Performance was strongly condition-dependent: delirium was near ceiling, whereas stroke remained hardest. Hourly sequence modeling improved AUROC on only three of seven eligible binary tasks. General-purpose models led the aggregate ranking over medical-domain adapted models. **Conclusions:** TIMELY Bench provides a reproducible pre-deployment instrument for testing when temporal structure helps, where models fail, and whether confidence and evidence grounding align with accuracy.

Keywords: critical care, temporal reasoning, multimodal clinical AI, benchmarking, large language models

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1 Introduction

ICU reasoning is temporal in a practical, bedside sense. Clinicians judge kidney injury, delirium, sepsis, and neurological deterioration from sequences of measurements, treatment changes, nursing observations, and notes, while tracking what was knowable at a particular decision time. A model can perform well on static prediction or medical knowledge recall and still fail at this partially observed trajectory problem. This distinction matters as generalist medical AI systems are proposed as multimodal assistants beyond single-task classifiers[1], and as reviews of biomedical LLMs continue to note the gap between broad language competence and clinically grounded evaluation[2, 3].

Existing critical care benchmarks have made ICU modeling reproducible but usually attach a prediction label to a structured window. MIMIC-III enabled large-scale ICU modeling[4]; MIMIC-Extract, the Harutyunyan multitask benchmark, Purushotham’s deep-learning comparison, and subsequent eICU and HiRID benchmarks standardized structured time-series prediction across mortality, decompensation, length-of-stay, and phenotype targets[5–9]. Earlier multimodal ICU studies and clinical language models added notes and reusable text representations[10, 11]. Longitudinal EHR foundation models and EHRSHOT further shifted attention to coded patient histories and transfer learning[12, 13]. These resources are foundational, but their evaluation unit is usually a stay, a window, or a patient-level label. They do not fix the same patient at the same anchor, mask future evidence, and ask whether a model can identify threshold crossing, trajectory change, or supporting evidence across synchronized EHR modalities.

This gap is not simply a matter of using longer context windows. A longitudinal model may learn useful patient-history representations without being tested on what could have been known at a particular ICU hour. Similarly, a multimodal prompt may contain structured measurements and notes without forcing the model to respect the temporal boundary between visible and future evidence. TIMELY Bench therefore treats the information state itself as part of the evaluation target.

Medical LLM benchmarks answer complementary questions. MedQA and PubMedQA test medical knowledge and biomedical reading comprehension[14, 15]. HealthBench, MedHELM, LLMEval-Med, and Med-Gemini extend evaluation toward open-ended and clinician-informed clinical tasks[16–19]. MedAgentBench evaluates tool use in a virtual EHR workflow[20]. These benchmarks are clinically important, but they rarely make the information boundary itself the object of evaluation. General temporal reasoning benchmarks such as TRAM and Test of Time test event ordering and temporal fact recall outside real patient trajectories[21, 22], while TIMER and EHR instruction-tuning work address training-time longitudinal reasoning[23, 24]. TIMELY Bench targets a narrower evaluation construct: anchor-bounded temporal clinical reasoning over real ICU trajectories.

The distinction is practical. A workflow benchmark asks whether an agent can retrieve and use information; a temporal reasoning benchmark asks whether the conclusion follows from the information that was actually visible at the anchor. Both are needed before clinical deployment, but they diagnose different failure modes.

Temporal reasoning depends not only on context length, but also on information state. The same stay can support different conclusions at hour 6, hour 24, and hour 48. A benchmark that leaks future evidence, collapses the trajectory into one global summary, or treats all conditions as equivalent cannot distinguish temporal reasoning from use of distal clues. This issue is especially acute in ICU data because the clinical record is not a neutral transcript of physiology: measurements are irregular, treatment changes alter the trajectory being observed, and notes may document suspicion before a formal label appears. We therefore framed TIMELY Bench around three requirements: same-stay/same-anchor synchronization across representations, explicit anchor-bounded visibility, and condition-specific physiologic structure.

TIMELY Bench places ICU stays on a shared 168-hour grid across four condition families: acute kidney injury (AKI), delirium, sepsis, and stroke. It materializes four synchronized views of the same anchor: Branch A, a compact structured summary; Branch B1, an hourly temporal sequence; Branch B2, a prompt-ready multimodal EHR context; and Branch B3, a state-space reconstruction. Evaluation uses the Clinical Reasoning Evaluation Suite (CRES), which decomposes temporal clinical reasoning into temporal localization (D1), threshold reasoning (D2), trend interpretation (D3), diagnostic discrimination (D4), contrastive reasoning (D5), and attribution (D6). For AKI, delirium, and sepsis, executable physiology templates define anchor events, phase windows, expected marker trajectories, and atypical variants; the frozen release applies these template rules to B3 trajectories to produce hourly phase labels and trajectory tiers.

This report asks three questions. First (RQ1), does preserving explicit temporal structure materially help relative to a compact summary representation? Second (RQ2), are some conditions more difficult for temporal reasoning than others? Third (RQ3), does medical-domain fine-tuning improve longitudinal multimodal reasoning, or mainly support static medical question answering? Its main contributions are: defining anchor-bounded temporal clinical reasoning as an evaluation target; synchronizing four EHR representations on a shared ICU grid; operationalizing CRES as a six-dimension reasoning framework; and evaluating structured baselines, nine LLM providers, and judge-rated response quality under a frozen protocol.

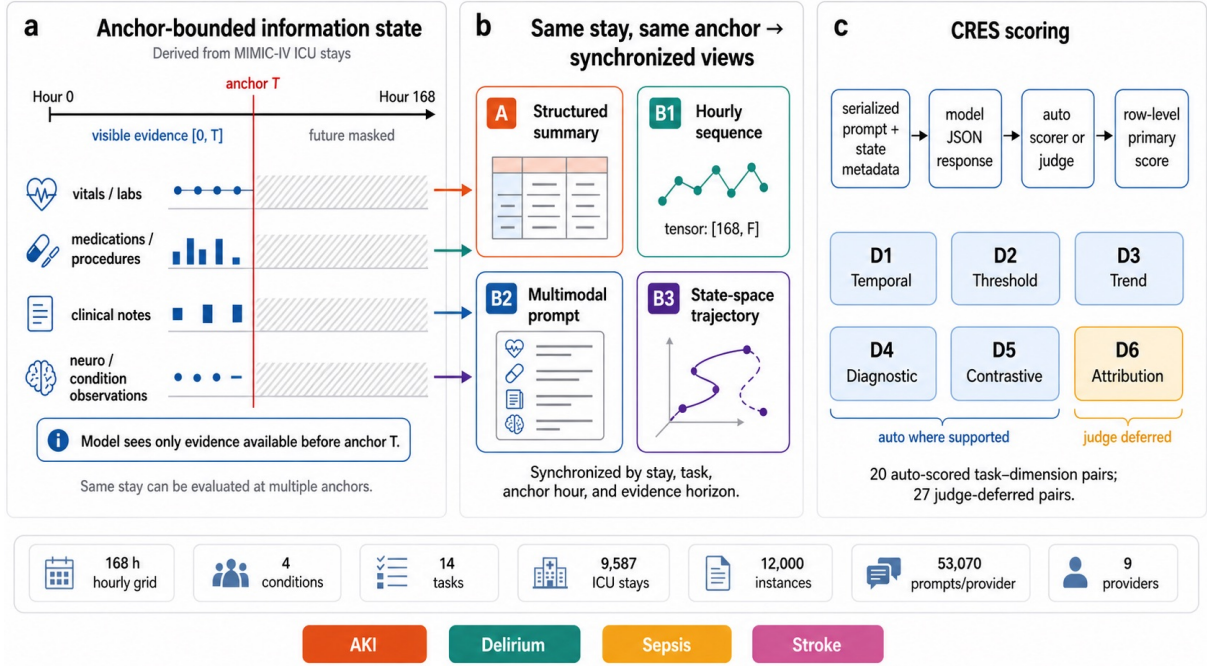


Fig. 1 TIMELY Bench evaluates temporal clinical reasoning by fixing the patient information state at an anchor hour and synchronizing multiple representations of the same ICU stay. Panel a shows that models can only use evidence visible in $[0, T]$, with future information masked. Panel b shows the four synchronized benchmark views: Branch A structured summary, Branch B1 hourly sequence, Branch B2 multimodal EHR prompt, and Branch B3 state-space trajectory. Panel c shows how a Clinical Reasoning Evaluation Suite (CRES) instance maps a stay, task, dimension, and anchor into a serialized prompt, JSON response, and scorer. The bottom scale strip summarizes the frozen evaluation scale.

2 Methods

2.1 Methodological overview

TIMELY Bench was constructed as a retrospective benchmark for temporal clinical reasoning over ICU data derived from MIMIC-IV v3.1 and MIMIC-IV Note[25]. Source tables were accessed through the credentialed PhysioNet BigQuery distribution, including ICU, hospital, derived, discharge-note, and radiology-note modules. The pipeline had six stages: extracting condition-specific ICU cohorts; aligning structured and text evidence to the first 168 ICU hours; defining anchor-bounded tasks for AKI, delirium, sepsis, and stroke; materializing synchronized representation branches; generating CRES prompt instances; and freezing canonical model outputs for scoring and comparison. The study is retrospective and methodologically aligned to STROBE, RECORD, and MI-CLAIM reporting domains[26–28].

The unit of analysis was not a patient, a whole stay, or a single prediction horizon. It was an anchor-defined information state. This design allowed multiple questions to be asked of the same stay at the same visible time point and made leakage control part of the benchmark specification rather than a post-hoc preprocessing choice.

The benchmark was anchor-bounded by design. Each prompt instance corresponds to one stay, one task, one reasoning dimension, and one anchor hour. Evidence visible to the model is truncated to ICU admission through that anchor, and future information is masked. This prevents temporal reasoning from being conflated with retrospective leakage. Prospective tasks in AKI, delirium, sepsis, and stroke Layer 1 use this rule. Stroke Layer 2 is explicitly retrospective

because mechanism, treatment strategy, NIHSS extraction, and complication labels depend on discharge or stay-level context.

This separation was important for validity. Prospective tasks test reasoning before the future outcome is visible, whereas stroke Layer 2 intentionally tests synthesis after the full clinical course has been documented. Treating those settings as one task family would mix prediction with retrospective abstraction.

This study used de-identified data from MIMIC-IV v3.1 and MIMIC-IV Note. Access requires completion of a recognized human subjects training course and a data use agreement. The MIMIC-IV dataset was collected under a waiver of informed consent approved by the Beth Israel Deaconess Medical Center institutional review board. No additional ethical approval was required for this secondary analysis of de-identified data.

2.2 Hourly alignment and synchronized representations

Heterogeneous EHR sources were mapped to hourly or event-time representations over the first 168 ICU hours, including structured observations, medications, procedures, diagnosis-pathway events, and note fragments. Four synchronized views were materialized for the same stay and anchor. Branch A is a compact structured summary for conventional baselines. Branch B1 preserves an explicit hourly sequence. Branch B2 is the prompt-ready multimodal EHR representation used for LLM evaluation; here, multimodal means EHR modalities, not imaging or audio. Branch B3 reconstructs a low-dimensional state-space trajectory for AKI, delirium, and sepsis.

The template layer applied condition-specific executable rules derived from physiology templates rather than inferring latent mechanisms from free text. These rules converted B3 trajectories into coarse phase labels and trajectory tiers using template-defined phase windows, expected marker trajectories, and atypical variants. Template support scores quantify observation coverage within phase-aligned windows; they do not prove that the full clinical narrative follows an idealized trajectory.

Template support is therefore a data-availability and templatability measure, not a hidden label of disease truth. A low-support stay can still be clinically valid; it simply lacks enough aligned observations for the executable template to assign a high-confidence phase pattern.

2.3 Condition definitions and label construction

AKI. AKI construction used the derived `kdigo_stages` table and retained stays with AKI evidence. Onset was anchored to the first hour meeting KDIGO Stage 1 criteria: creatinine increase of at least 0.3 mg/dL within 48 hours, creatinine at least 1.5 times baseline, or urine output below 0.5 mL/kg/h for at least 6 hours[29]. Stage 1, Stage 2+, Stage 3, and renal replacement therapy (RRT) proxy timing were extracted from derived tables, procedure events, and hourly RRT indicators. Anchors were emitted every 4 hours after Stage 1 onset. AKI-T1 is positive if

Stage 2+ onset falls in $(T, T + 24]$; AKI-S1 is positive if an RRT proxy event falls in $(T, T + 72]$. Post-progression anchors were excluded so prompts remained pre-escalation information states. **Delirium.** Delirium onset was anchored to the first positive CAM-ICU assessment from charted delirium assessments (itemid 228332), requiring the patient to be assessable at RASS ≥ -3 [30]. The template approximated DSM-5 delirium constructs using executable hourly ICU observations rather than bedside neuropsychiatric narratives[31]. DEL-T1 captures persistent delirium within $(T, T + 24]$; DEL-S1 captures resolution within $(T, T + 24]$ using a sustained 24-hour non-positive window. Labels retain `left_censored`, delirium-hour counts, and support flags for RASS, GCS, and restraints to separate physiology from observation sparsity. The label is therefore a structured screening-derived proxy, not an expert chart-adjudicated psychiatric diagnosis.

Sepsis. Sepsis construction started from stays flagged `sepsis3` and derived onset timing from diagnosis-pathway events. Explicit sepsis-onset events defined the onset hour when available; otherwise onset was set to hour zero and marked `onset_confidence=low`. Septic shock was operationalized as mean arterial pressure below 65 mmHg with active vasopressor support, with lactate > 2 mmol/L as an additional gate when available, consistent with Sepsis-3 and Surviving Sepsis Campaign guidance[32, 33]. SEP-T1 is positive when first post-onset shock falls in $(T, T + 12]$; SEP-S1 is positive when lactate in $(T, T + 6]$ falls to at most 90% of observed baseline.

Sepsis tasks were designed around deterioration after a sepsis information state, not around discovering sepsis de novo from the entire record. The `onset_confidence` field allows analyses to separate explicit onset timing from lower-confidence pathway-derived defaults.

Stroke. Stroke used a stricter phenotype filter: only stays with `stroke_subtype_priority == "ischaemic"` entered the main benchmark, consistent with acute ischaemic stroke pathways[34]. Layer 1 contains anchor-bounded temporal tasks over motor strength, affected side, imaging consistency, and sequence signatures. S-T1 asks whether limb strength worsens in the next 12 hours; S-T2 derives affected side; S-T3 compares neurological laterality with first brain imaging; and S-T4 summarizes the temporal ordering of assessment, pathway activation, imaging, and first worsening. Layer 2 contains retrospective mechanism, strategy, NIHSS extraction, and complication tasks from discharge summaries and stay-level context.

The two-layer design was retained because stroke combines short-horizon neurological evolution with retrospective clinical synthesis. Pooling the two would make stroke appear artificially inconsistent: the same condition contains both prediction tasks under future masking and tasks that intentionally use the full documented course.

2.4 Prompt serialization, baselines, and model outputs

The formal LLM evaluation used Branch B2 and a fixed “Modality-Separated Sections” serialization. Prompts begin with a system instruction and patient-context header naming the

condition, task, and anchor hour, followed by sections for demographics, structured hourly timeline, medications, procedures, and clinical notes; stroke temporal prompts additionally include neurological assessments. Each prompt ends with a task-specific question and a fixed JSON schema containing reasoning, answer, timestamped evidence, and confidence. If a value was absent from a rendered hour, it means the value was not available in that hour’s visible evidence stream, not that it was imputed as normal or zero. An illustrative AKI-T1 prompt and two model responses are provided in Appendix A.

Five prompt variants were materialized, but formal comparative results use only the frozen `full_multimodal` variant. Structured baselines were evaluated only on eligible binary tasks. Branch A used XGBoost on 384 engineered anchor-state features. Branch B1 used BiLSTM-attention and temporal-transformer models on fixed-length 168-hour structured sequences aligned to the same anchor-bounded evidence state. Missingness was represented by released masks and eligibility fields, and all baselines used five-fold subject-level cross-validation. Figure 2 uses the B1 model selected by AUROC within each task as the shared B1 comparator for both AUROC and AUPRC deltas.

This baseline design separates two questions. Branch A asks whether engineered anchor-state summaries are sufficient for the eligible binary targets. Branch B1 asks whether preserving the full hourly structured sequence adds useful information once patient-level leakage is controlled by subject-level cross-validation.

The frozen comparative set contained nine providers: GPT-5.4, Gemini 3.1 Pro, DeepSeek Chat, Qwen3.5, Gemma4-26B, Aloe-70B, Aloe-7B, Meditron3-8B, and MedGemma-4B. Tier 1A denoted closed general-purpose API models, Tier 1B general-purpose open or semi-open models, and Tier 2 medical-domain adapted models. Canonicalization required one response file per provider with 53,070 unique prompt identifiers, `status=ok`, and `parse_success=true`. Provider-specific decoding defaults, output-token limits, and repair limits were not assumed identical; release manifests record provenance, row counts, token totals, latency summaries, and repair status.

2.5 CRES dimensions, scoring, and judge evaluation

The Clinical Reasoning Evaluation Suite (CRES) decomposes temporal clinical reasoning into six dimensions. D1 temporal localization asks when the earliest sign of the target event became visible. D2 threshold reasoning asks whether a clinically meaningful threshold was crossed and, if so, when. D3 trend interpretation asks whether the relevant signal is improving, stable, or worsening. D4 diagnostic discrimination asks which explanation or mechanistic category best fits the evidence. D5 contrastive reasoning asks the model to choose between competing trajectory types or support classes. D6 attribution asks the model to identify the evidence items that most strongly support the answer.

Operationalization is task-specific rather than one-size-fits-all. The generalized auto-scoring pipeline reused the same parser, truth builder, and scoring rules originally validated on Tier 1A.

Primary score was defined as binary accuracy for event-time and binary tasks, macro F1 for categorical tasks, and exact match for numeric extraction. Provider-level macro primary score was computed as the equal-weight mean of the 20 supported task-dimension primary scores for each provider, not as a row-weighted mean. D6 remained deferred from primary auto-scoring and was assessed in the judge phase.

For event-time tasks, the scorer first normalizes abstentions, no-event answers, and explicit hour values before comparing them with the task truth object. Thus an answer can be correct either by identifying the correct event state or by correctly abstaining when the visible anchor-bounded evidence does not support an event within the target horizon.

Judge evaluation was built after the auto-scored comparative set froze. The packet contained 500 prompt instances, sampled as 125 per condition, with 60% from deferred pairs and 40% from hard auto-scorable disagreements. The fixed contestant roster was GPT-5.4, DeepSeek Chat, Aloe-70B, and MedGemma-4B, yielding 2,000 judged responses. Claude Opus 4.6 served as primary judge, with GPT-5.4 and Gemini 3.1 Pro as cross-check judges scoring the same packet, following general LLM-as-judge methodology adapted to structured clinical reasoning[35]. CREATE was used to construct frozen scoring artifacts and the judge packet; final external judge execution, repair, and aggregation were completed in the synchronized local analysis workspace after the CREATE-side Claude API run was blocked by a provider-side Cloudflare response.

Box 1 | Operational definition of the six Clinical Reasoning Evaluation Suite (CRES) dimensions. Supported auto-scored pairs are evaluated in Phase 6.5F; D6 attribution remains judge-deferred in the primary results.

Dimension	Clinical question type	Truth construction in TIMELY Bench	Primary score	Formal status
D1 Temporal localization	When did the earliest sign of the target event become visible?	Earliest post-anchor onset hour from task tables, such as Stage 2+ onset, delirium onset, sepsis onset, or first neurological worsening hour.	Binary accuracy on event-time detection	Auto-scored
D2 Threshold reasoning	Has a clinically meaningful threshold been crossed, and if so when?	Guideline-linked or task-linked threshold hour, such as KDIGO stage crossing, RRT proxy timing, septic shock onset, or peak NIHSS value.	Binary accuracy or exact match	Auto-scored
D3 Trend interpretation	Is the relevant physiological signal improving, stable, or worsening?	Future trajectory label derived from structured follow-up windows, most prominently stroke motor-strength worsening and related temporal tasks.	Binary accuracy	Auto-scored where supported
D4 Diagnostic discrimination	Which clinical explanation best fits the observed evidence?	Categorical truth from stay-level labels such as affected side, imaging-consistency class, stroke mechanism, treatment strategy, or complication presence.	Macro F1	Auto-scored
D5 Contrastive reasoning	Which competing trajectory archetype best matches the case?	Condition-specific contrast labels, including AKI typical versus atypical trajectory class and sepsis onset-confidence class.	Macro F1	Auto-scored where supported
D6 Attribution	Which evidence items most strongly support the answer?	Judge-facing evidence grounding over submitted timestamped evidence triples and free-text rationale.	Rubric-based judge scoring	Deferred

3 Results

Results are organized around the three research questions and then by diagnostic analyses that explain model failure modes. Four findings guide the section: Branch B1 improved AUROC on only three of seven eligible binary tasks; general-purpose models led the aggregate provider ranking; CRES showed that trend reasoning was stronger than diagnostic discrimination; and trajectory-template support and the stroke layer split exposed distinct difficulty regimes.

3.1 Dataset scale and benchmark composition

TIMELY Bench is a multi-condition benchmark rather than a repeated binary prediction template. The full release covered 2,456,896 AKI task instances from 49,920 ICU stays, 1,512,502 delirium instances from 21,628 stays, 898,560 sepsis instances from 29,713 stays, and 61,111 stroke instances from 5,318 stays (Table 1). AKI, delirium, and sepsis each contributed two onset- or state-oriented tasks on a common temporal scaffold; stroke contributed eight tasks split between Layer 1 temporal prediction and Layer 2 retrospective or extraction tasks (Table 1).

Representation availability differed by condition family. Branches A, B1, and B2 were available for all four conditions, whereas B3 and the physiology-template layer were available only for AKI, delirium, and sepsis. Stroke retained anchor-bounded multimodal prompts but was not forced into a phase-template stack because its progression signals were organized around neurological deficit structure, laterality, imaging concordance, and retrospective discharge synthesis.

Task balance also varied. Positive rates ranged from 0.98% for AKI-S1 and 1.66% for SEP-T1 to approximately 21–22% for the delirium tasks; stroke combined binary, categorical, and numeric extraction settings. The frozen formal sample contained 12,000 benchmark instances from 9,587 ICU stays, capped at two anchors per stay per task where applicable. Expanding these instances across the supported task-dimension map yielded 53,070 `full_multimodal` prompts per provider.

Prompt length was clinically variable but feasible for the frozen provider set. The median `full_multimodal` prompt contained approximately 1,724 tokens (interquartile range 807–2,564; range 183–15,040), with a mean of 1,978 tokens. Stroke prompts were longest on average (median 2,359 tokens) and sepsis shortest (1,120). Clinical notes contributed 58.7% of prompt tokens on average, followed by structured timeline and medication sections.

The length distribution also reflects the task design. Stroke retrospective tasks had the longest prompts because they drew on discharge and stay-level context, whereas some anchor-bounded stroke laterality tasks were much shorter. The benchmark therefore includes both compact temporal classification prompts and longer multimodal synthesis prompts, but all formal comparisons used the same frozen prompt format.

Table 1 Dataset statistics. Panel A summarizes condition-level cohort size, task count, task roster, and representation availability.

Condition	Instances	Unique stays	Tasks	Representations	B3
AKI	2,456,896	49,920	AKI-S1, AKI-T1	A/B1/B2/B3	yes
Delirium	1,512,502	21,628	DEL-S1, DEL-T1	A/B1/B2/B3	yes
Sepsis	898,560	29,713	SEP-S1, SEP-T1	A/B1/B2/B3	yes
Stroke	61,111	5,318	S-R1, S-R2, S-R3, S-R4, S-T1, S-T2, S-T3, S-T4	A/B1/B2	no

Table 1: Dataset statistics. Panel B summarizes task-level instance counts, unique stays, label summary, and representation availability. Disease prefixes denote AKI, delirium (DEL), sepsis (SEP), and stroke (S); task-family letters after the hyphen denote temporal or prospective tasks (T), secondary state or trajectory tasks (S), and retrospective stroke extraction or synthesis tasks (R).

Task	Condition	Layer / mode	Instances	Unique stays	Label summary	Representations
AKI-S1	AKI	single-layer temporal	1,764,883	48,461	0.98%	A/B1/B2/B3
AKI-T1	AKI	single-layer temporal	692,013	40,915	8.46%	A/B1/B2/B3
DEL-S1	Delirium	single-layer temporal	756,251	21,628	22.03%	A/B1/B2/B3
DEL-T1	Delirium	single-layer temporal	756,251	21,628	21.10%	A/B1/B2/B3
SEP-S1	Sepsis	single-layer temporal	27,003	11,715	21.11%	A/B1/B2/B3
SEP-T1	Sepsis	single-layer temporal	871,557	26,517	1.66%	A/B1/B2/B3
S-R1	Stroke	Layer 2 retrospective	3,912	3,912	5 classes	B2
S-R2	Stroke	Layer 2 retrospective	3,912	3,912	4 classes	B2
S-R3	Stroke	Layer 2 retrospective	277	277	median=2.5	B2
S-R4	Stroke	Layer 2 retrospective	3,912	3,912	8.79%	B2
S-T1	Stroke	Layer 1 temporal	36,519	4,792	14.25%	A/B1/B2
S-T2	Stroke	Layer 1 temporal	4,952	4,952	5 classes	A/B1/B2
S-T3	Stroke	Layer 1 temporal	2,651	2,651	3 classes	A/B1/B2
S-T4	Stroke	Layer 1 temporal	4,976	4,976	16 classes	A/B1/B2

3.2 Explicit temporal structure helped selectively

The branch comparison tested RQ1 by asking whether an hourly structured sequence improved over a compact anchor-state summary in conventional baselines. Branch A XGBoost was strong for AKI-S1 (AUROC 0.977), AKI-T1 (0.941), both delirium tasks (0.918 and 0.916), and SEP-T1 (0.950), but weaker for SEP-S1 (0.751) and S-T1 (0.726) (Table 2).

Explicit hourly sequence modeling did not uniformly dominate. Using the AUROC-selected B1 model for each task as the shared comparator, B1 improved AUROC on only three of seven eligible tasks: AKI-S1, AKI-T1, and SEP-T1. It improved AUPRC on only one task, AKI-T1, where BiLSTM-attention increased AUROC from 0.941 to 0.949 and AUPRC from 0.595 to 0.603. The largest B1 deficit was SEP-S1, where B1 underperformed A by 0.047 AUROC and 0.057 AUPRC. Temporal structure helped selectively, not automatically.

This result is important because the branch comparison is not an LLM prompt ablation. It isolates the structured-data question of whether preserving the hourly sequence improves over summarizing the anchor state. The answer was task-dependent: sequence models helped when evolving physiology was recoverable from the hourly grid, but not when the summary already captured the signal or when the sequence introduced sparse, noisy observations.

This finding is also a useful negative control for the rest of the report. If explicit sequence models had dominated every task, later LLM failures could be interpreted mainly as prompt

Table 2 Structured baselines on eligible binary tasks. Five-fold subject-level cross-validation results for Branch A and Branch B1, reported as mean AUROC and AUPRC.

Task	Condition	Branch	Model	AUROC	AUPRC
AKI-S1	AKI	A	xgboost	0.977	0.372
AKI-S1	AKI	B1	temporal_transformer	0.978	0.346
AKI-S1	AKI	B1	bilstm_attention	0.977	0.339
AKI-T1	AKI	A	xgboost	0.941	0.595
AKI-T1	AKI	B1	bilstm_attention	0.949	0.603
AKI-T1	AKI	B1	temporal_transformer	0.941	0.562
DEL-S1	Delirium	A	xgboost	0.918	0.702
DEL-S1	Delirium	B1	temporal_transformer	0.911	0.689
DEL-S1	Delirium	B1	bilstm_attention	0.910	0.702
DEL-T1	Delirium	A	xgboost	0.916	0.692
DEL-T1	Delirium	B1	bilstm_attention	0.908	0.659
DEL-T1	Delirium	B1	temporal_transformer	0.894	0.615
SEP-S1	Sepsis	A	xgboost	0.751	0.465
SEP-S1	Sepsis	B1	bilstm_attention	0.704	0.408
SEP-S1	Sepsis	B1	temporal_transformer	0.643	0.338
SEP-T1	Sepsis	A	xgboost	0.950	0.329
SEP-T1	Sepsis	B1	bilstm_attention	0.953	0.275
SEP-T1	Sepsis	B1	temporal_transformer	0.948	0.258
S-T1	Stroke	A	xgboost	0.726	0.251
S-T1	Stroke	B1	bilstm_attention	0.707	0.247
S-T1	Stroke	B1	temporal_transformer	0.676	0.230

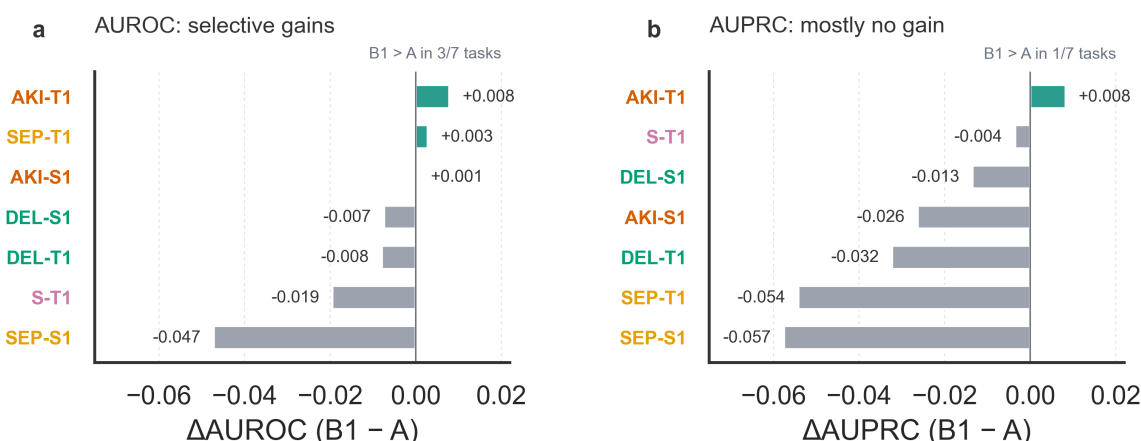


Fig. 2 Explicit hourly sequence modeling improved performance only selectively, indicating that the value of temporal detail was task-dependent. Panel a shows delta AUROC and panel b shows delta AUPRC, both computed as Branch B1 minus Branch A for the seven eligible binary tasks. For each task, Branch B1 is represented by the B1 model selected by AUROC, and both AUROC and AUPRC deltas are computed using that same selected B1 comparator relative to Branch A. B1 improved AUROC in 3/7 tasks (AKI-S1, AKI-T1, SEP-T1) and improved AUPRC in 1/7 tasks (AKI-T1 only). Negative deltas indicate tasks where the simpler structured baseline outperformed the explicit hourly-sequence model.

or model limitations. Instead, the structured baselines show that temporal detail itself is conditional: the representation helps only when the task-relevant signal is both temporal and measurable in the available stream.

3.3 General-purpose models led the aggregate ranking

The provider comparison tested RQ3 by evaluating whether medical-domain adaptation improved anchor-bounded temporal reasoning. The formal set included nine frozen providers: two Tier 1A closed general-purpose API models, three Tier 1B general-purpose open or semi-open models, and four Tier 2 medical-domain adapted models. Tier labels were analysis strata

Table 3 Frozen cross-provider comparative ranking across the nine-provider formal comparative set. Pair wins denote the number of supported task-dimension pairs for which the provider achieved the highest primary score within its tier; ties are counted for all co-winning providers.

Provider	Tier	Macro score	Pair wins	Avg latency (s)	Total tokens
Gemini 3.1 Pro	Tier 1A	0.655	16	15.938	343,643,633
Gemma4-26B	Tier 1B	0.646	12	30.038	277,833,535
DeepSeek Chat	Tier 1B	0.635	10	12.711	261,196,668
Qwen3.5	Tier 1B	0.634	4	4.843	285,969,716
GPT-5.4	Tier 1A	0.626	9	11.401	264,064,840
MedGemma-4B	Tier 2	0.535	8	71.572	285,394,344
Aloe-70B	Tier 2	0.519	11	5.710	258,039,766
Meditron3-8B	Tier 2	0.510	7	3.198	259,948,539
Aloe-7B	Tier 2	0.489	7	2.759	282,111,131

rather than performance-derived categories. Macro primary score was the equal-weight mean over 20 supported task-dimension scores, not a row-weighted average.

Overall macro primary score ranged from 0.489 to 0.655 (Table 3). Gemini 3.1 Pro led at 0.655, followed by Gemma4-26B (0.646), DeepSeek Chat (0.635), Qwen3.5 (0.634), and GPT-5.4 (0.626). The best Tier 2 model, MedGemma-4B, reached 0.535, below all Tier 1A and Tier 1B providers. Aloe-70B, Meditron3-8B, and Aloe-7B followed at 0.519, 0.510, and 0.489.

Condition-level summaries explain why overall averages should be read cautiously (Table 4; Fig. 3). Delirium was near ceiling for several providers, AKI favored GPT-5.4 and Gemini 3.1 Pro, sepsis favored several Tier 1B providers, and stroke remained difficult for all tiers. Temporal clinical reasoning therefore did not behave as one uniform capability.

The condition pattern also prevents an over-simple interpretation of the tier gap. Tier 2 models did not fail uniformly on every slice, and some condition-level comparisons were close. The robust statement is that the tested medical-domain adapted models did not outperform the general-purpose groups in aggregate or at the leading-provider level under this frozen evaluation design.

Table 4: Condition-level comparative summary. Condition-stratified provider performance summarized as macro primary score within each condition family.

Provider	Tier	Condition	Macro score	Pairs
GPT-5.4	Tier 1A	AKI	0.789	5
Gemini 3.1 Pro	Tier 1A	AKI	0.786	5
DeepSeek Chat	Tier 1B	AKI	0.753	5
Gemma4-26B	Tier 1B	AKI	0.708	5
Qwen3.5	Tier 1B	AKI	0.679	5
MedGemma-4B	Tier 2	AKI	0.477	5
Aloe-70B	Tier 2	AKI	0.458	5
Meditron3-8B	Tier 2	AKI	0.418	5
Aloe-7B	Tier 2	AKI	0.342	5
DeepSeek Chat	Tier 1B	Delirium	1.000	3
Gemini 3.1 Pro	Tier 1A	Delirium	1.000	3
GPT-5.4	Tier 1A	Delirium	1.000	3
Aloe-70B	Tier 2	Delirium	1.000	3
Gemma4-26B	Tier 1B	Delirium	0.996	3
Qwen3.5	Tier 1B	Delirium	0.993	3
Aloe-7B	Tier 2	Delirium	0.964	3
Meditron3-8B	Tier 2	Delirium	0.958	3
MedGemma-4B	Tier 2	Delirium	0.883	3
Gemma4-26B	Tier 1B	Sepsis	0.833	4
Qwen3.5	Tier 1B	Sepsis	0.822	4
Gemini 3.1 Pro	Tier 1A	Sepsis	0.802	4
DeepSeek Chat	Tier 1B	Sepsis	0.795	4
GPT-5.4	Tier 1A	Sepsis	0.750	4
MedGemma-4B	Tier 2	Sepsis	0.714	4
Meditron3-8B	Tier 2	Sepsis	0.686	4

Provider	Tier	Condition	Macro score	Pairs
Aloe-7B	Tier 2	Sepsis	0.632	4
Aloe-70B	Tier 2	Sepsis	0.556	4
Gemma4-26B	Tier 1B	Stroke	0.382	8
Qwen3.5	Tier 1B	Stroke	0.376	8
Gemini 3.1 Pro	Tier 1A	Stroke	0.371	8
Aloe-70B	Tier 2	Stroke	0.359	8
MedGemma-4B	Tier 2	Stroke	0.350	8
DeepSeek Chat	Tier 1B	Stroke	0.344	8
Aloe-7B	Tier 2	Stroke	0.331	8
GPT-5.4	Tier 1A	Stroke	0.321	8
Meditron3-8B	Tier 2	Stroke	0.312	8



Fig. 3 General-purpose models led the frozen comparison, while Clinical Reasoning Evaluation Suite (CRES) stratification showed dimension-specific failure patterns. Panel a shows overall macro primary score by provider, with bars colored by model tier. Panel b shows condition-level macro primary scores across AKI, delirium, sepsis, and stroke. Panel c shows mean primary score by CRES dimension and provider tier for auto-scored dimensions D1–D5. D1–D5 are auto-scored where supported; D6 attribution is judge-deferred and excluded from primary auto-scoring.

3.4 Per-dimension CRES analysis exposed distinct reasoning failure modes

CRES separated aggregate performance into clinically interpretable operations (Box 1). Formal auto-scoring covered 20 supported task-dimension pairs across D1–D5; 27 pairs and D6 attribution were deferred to judge evaluation or qualitative inspection.

Across providers, D3 trend reasoning was strongest (mean primary score 0.797), followed by D1 temporal localization (0.762) and D2 threshold reasoning (0.702). D5 contrastive reasoning was lower (0.406), and D4 diagnostic discrimination was weakest (0.242). This ordering is clinically plausible: trend classification can be straightforward once the relevant trajectory is visible, whereas diagnostic discrimination requires choosing among overlapping explanations under partial evidence.

The largest tier separation appeared in D2. Tier 1A and Tier 1B both averaged approximately 0.881 on threshold reasoning, whereas Tier 2 averaged 0.478. The pattern was not uniform weakness: Tier 2 was competitive on D3 trend reasoning (0.802) and comparatively stronger

on D5. Condition-level dimension results reinforced this interpretation. D1 was near ceiling for delirium and sepsis, moderate for AKI, and near floor for stroke; D2 was high for delirium, intermediate for stroke and AKI, and lower for sepsis, where the target combines MAP, vasopressor exposure, lactate availability, and onset chronology.

Per-dimension reporting is therefore not a cosmetic decomposition of the leaderboard. It identifies which clinical operation fails. A model that can classify a visible trend may still miss a guideline-like threshold, and a model that localizes an event may still misattribute the clinical mechanism. This is why TIMELY Bench reports CRES dimensions alongside task and condition summaries.

The D4 result is especially important for interpretation. Diagnostic discrimination was not simply another binary label; it required models to choose among competing explanations under partial evidence. Low D4 performance therefore indicates a limitation in mechanistic interpretation, not just a failure to extract a number or repeat a threshold rule.

3.5 Confidence behavior varied independently of raw accuracy

Confidence behavior separated providers in a way that primary score alone did not. GPT-5.4 marked 37,086 of 53,070 outputs as high confidence (69.9%), whereas Gemini 3.1 Pro marked 52,635 outputs as high (99.2%). DeepSeek Chat was closer to GPT-5.4 at 71.6%; Qwen3.5 and Gemma4-26B were more compressed at 90.7% and 94.9%. Gemini 3.1 Pro led the overall score but used the high-confidence label almost uniformly, making uncertainty-aware triage or selective escalation harder to infer from the response schema.

This should not be read as probability calibration, because the benchmark used a categorical self-reported confidence field rather than calibrated risk estimates. It is still operationally relevant. A model that uses the same confidence label for nearly every response gives downstream reviewers less information about which cases may require escalation, audit, or abstention.

3.6 Temporal sensitivity varied by condition

The temporal stress test examined whether condition difficulty changed across early, middle, and later anchor windows.

Temporal degradation analysis used predefined anchor buckets instead of exact single-hour points: T6 for anchor hours 4–8, T24 for hours 18–30, and T48 for hours 46–50. Any provider-task-bucket cell with fewer than 25 instances was marked as insufficient support and excluded from the main plot. Delirium and stroke do not show a T24 marker in the main figure because the middle bucket failed the support threshold, not because the data were missing.

The temporal pattern was not monotonic (Fig. 4). Sepsis displayed a T24 peak before declining again at T48. A plausible explanation is signal concentration: the T24 window disproportionately captures high-information segments around sepsis-to-shock evolution, whereas T48 includes more heterogeneous and partially stabilized stays. The result is more specific

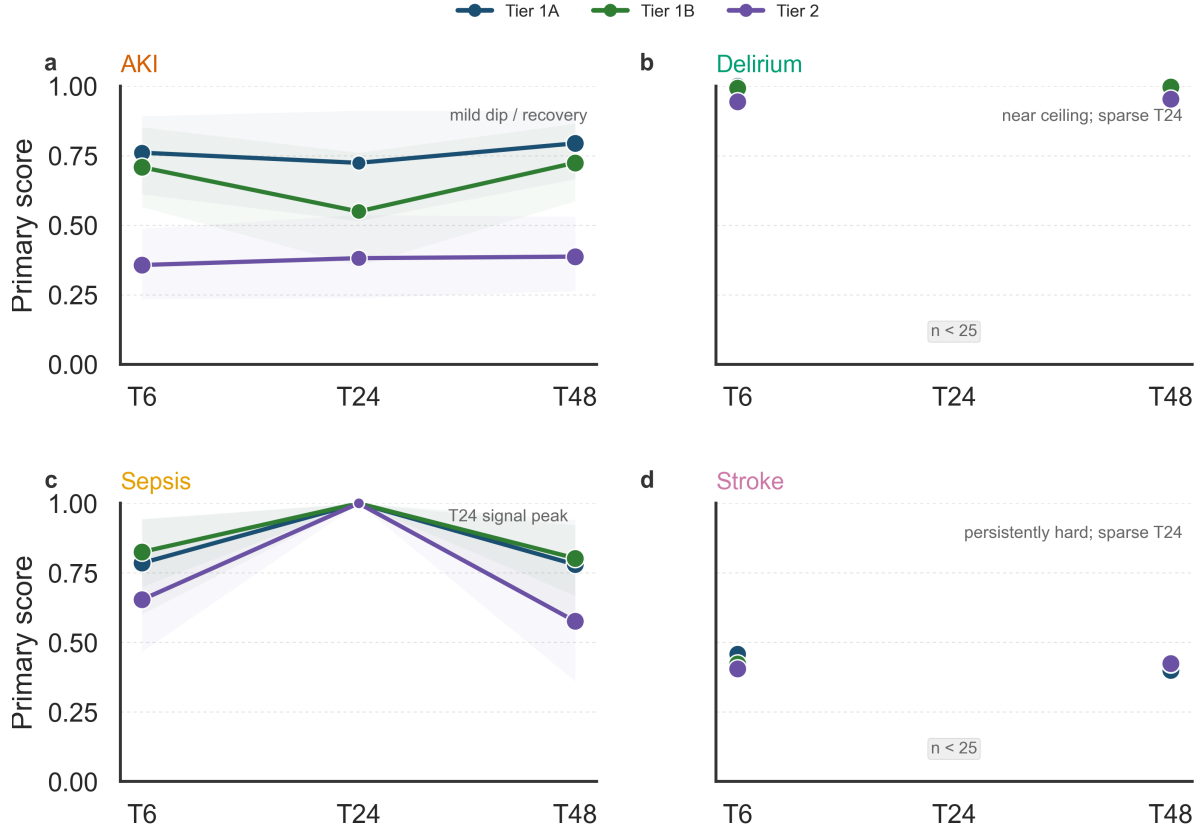


Fig. 4 Temporal sensitivity varied by condition and was not uniformly monotonic. Mean primary score is shown across predefined temporal buckets T6 (anchor hours 4–8), T24 (18–30), and T48 (46–50), stratified by provider tier and condition. Point size reflects the number of scored instances in each tier-condition-bucket cell, and shaded bands indicate bootstrap uncertainty where available. Cells with fewer than 25 instances were marked as insufficient support rather than treated as missing data. Sepsis showed a T24 signal peak, whereas delirium remained near ceiling and stroke remained persistently difficult.

than “earlier is harder” or “later is easier.” Anchor sensitivity depends on where clinically informative signal is concentrated along the trajectory.

3.7 The stroke two-layer split captured a real reasoning boundary

The stroke analysis separated prospective temporal prediction from retrospective synthesis within the same condition family.

Stroke was designed differently from the other three conditions because it contains two qualitatively distinct reasoning regimes. Layer 1 evaluates anchor-bounded temporal prediction over evolving neurological observations. Layer 2 evaluates retrospective mechanism, strategy, numeric extraction, and complication reasoning using discharge and stay-level context. The empirical results support this split.

Layer 2 outperformed Layer 1 for all nine providers, with a mean Layer 2 minus Layer 1 gain of 0.130 macro primary score (Fig. 5). The largest gains were observed for MedGemma-4B (+0.240), Gemma4-26B (+0.168), and Aloe-7B (+0.167). Retrospective synthesis from the full documented course was easier than short-horizon temporal stroke prediction under

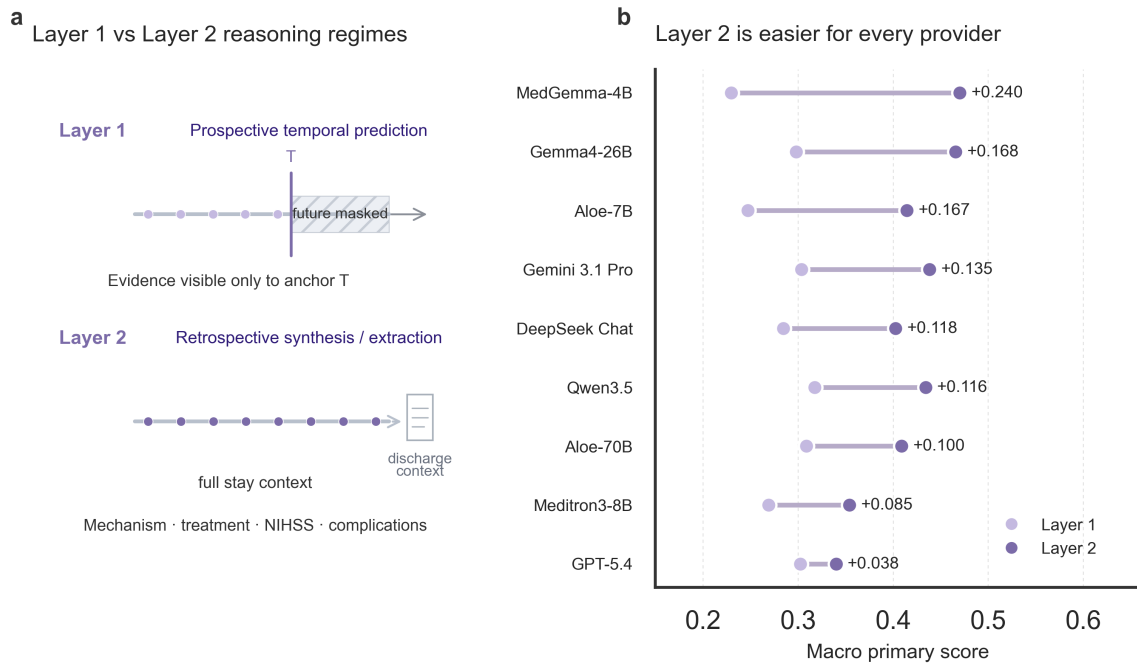


Fig. 5 The stroke two-layer split captured a real reasoning boundary: retrospective synthesis and extraction were consistently easier than anchor-bounded temporal prediction. Panel a contrasts Layer 1, where only pre-anchor evidence is visible and the future neurological state is masked, with Layer 2, where full stay-level and discharge context support retrospective mechanism, treatment, NIHSS, and complication extraction. Panel b shows provider-level macro primary score for Layer 1 and Layer 2; Layer 2 outperformed Layer 1 for all nine providers. The mean Layer 2 minus Layer 1 gain was approximately +0.130 macro primary score.

anchor masking. The same clinical story becomes much harder when the model must judge deterioration from partial evolving evidence, which is why stroke was separated into two layers.

3.8 Template support revealed a strong trajectory gradient

The trajectory-tier analysis tested whether physiology-template support identified clinically meaningful differences in trajectory difficulty.

Trajectory-tier analysis linked model performance to how well a stay matched an executable physiology template. Stay-level template-support coverage was 6% for AKI, 96% for delirium, and 37% for sepsis (Fig. 6, panel a). These are not model-performance values; they quantify whether adequate observations were available for template-aligned phase rules.

Row-level performance varied across tiers (Fig. 6, panel c). The gradient was strongest for delirium, intermediate for sepsis, and weakest for AKI. AKI also showed a counterintuitive pattern: low-support rows outperformed typical-supported rows, probably because low-support AKI contained more easy non-progression cases while typical-supported rows concentrated clinically dynamic trajectories. The important point is that template support exposes case-mix structure hidden by pooled averages.

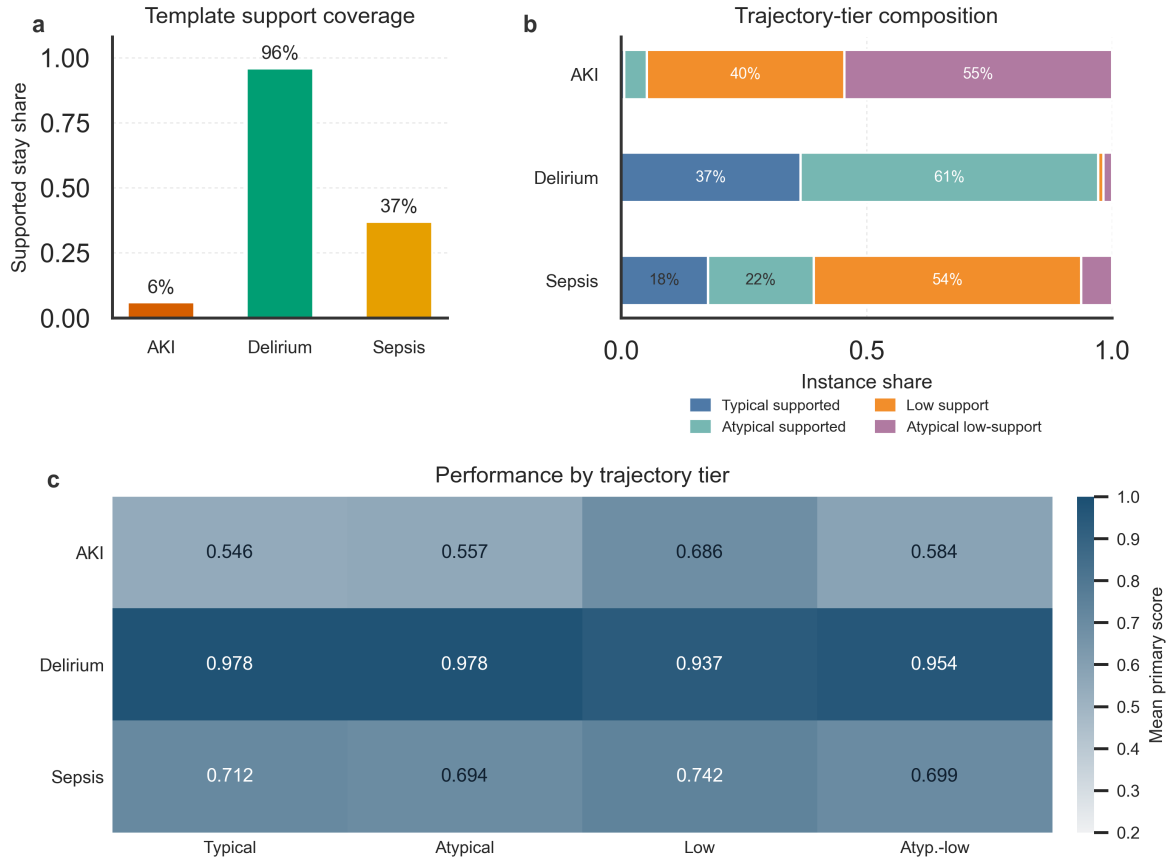


Fig. 6 Template support revealed disease-specific trajectory regularity and exposed performance differences hidden in pooled averages. Panel a shows stay-level template support coverage for AKI, delirium, and sepsis. Panel b shows sample-instance composition across trajectory tiers; only major composition segments are labeled. Panel c shows row-level auto-scored mean primary score by trajectory tier. Template support is an evidence-support / templatability measure, not a model-performance metric. The higher score in the AKI low-support stratum likely reflects easier non-progression case mix rather than a universal benefit of low support.

3.9 Judge-based evaluation preserved the auto-scored provider ordering

Judge evaluation tested whether the auto-scored ordering persisted on deferred and hard-disagreement cases. The fixed roster included GPT-5.4, DeepSeek Chat, Aloe-70B, and MedGemma-4B. The packet contained 500 prompt instances and 2,000 contestant responses, scored by Claude Opus 4.6, GPT-5.4, and Gemini 3.1 Pro on the same rows.

The judge summaries preserved the broad provider separation seen in auto-scoring (Table 5; Fig. 7). Scores were averaged across the three judges. GPT-5.4 led on all five rubric axes, followed by DeepSeek Chat, Aloe-70B, and MedGemma-4B. Cross-judge agreement was substantial: against Claude Opus 4.6, GPT-5.4 achieved Spearman ρ 0.814 on overall quality and 0.808 on clinical correctness, while Gemini 3.1 Pro achieved 0.783 and 0.759. The judge phase therefore served as a structured complementary evaluation layer, not a substitute for human adjudication.

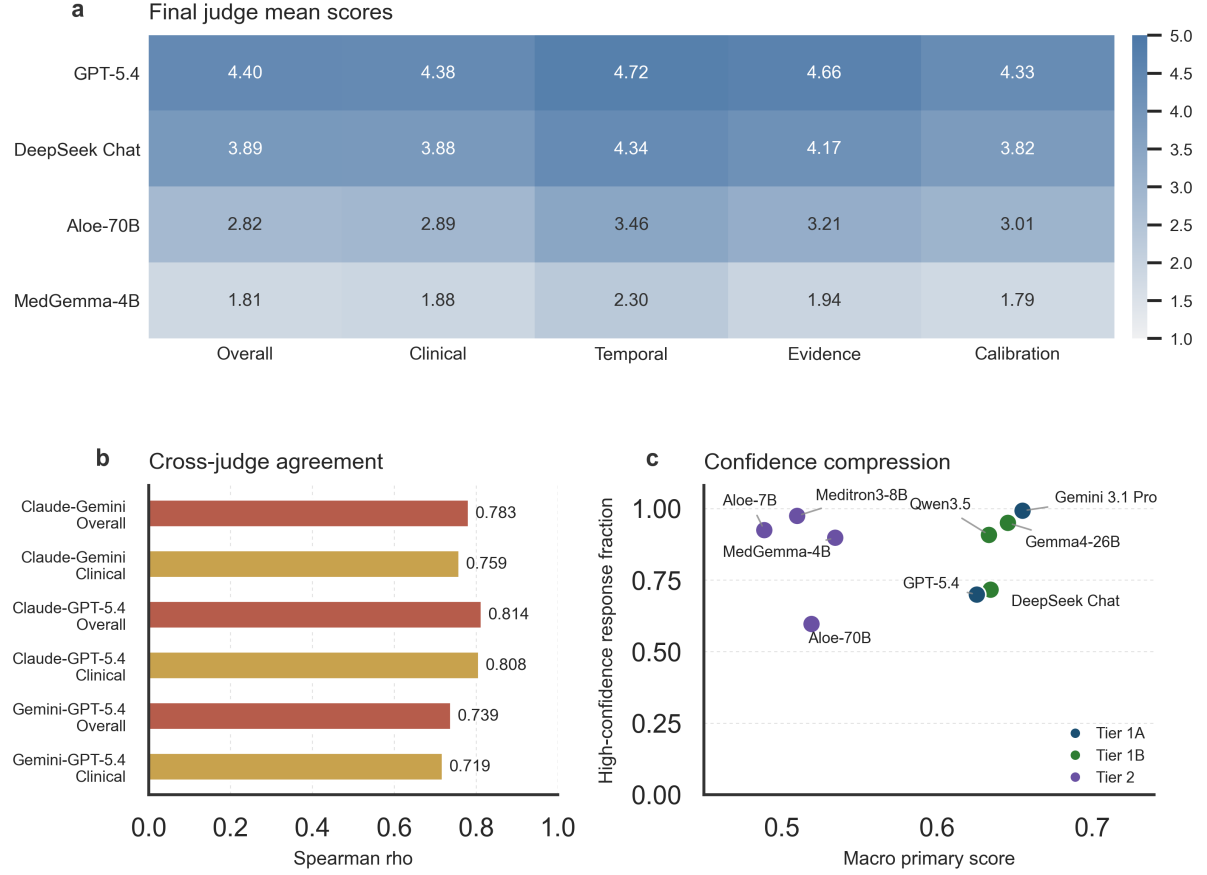


Fig. 7 Judge-based evaluation preserved the broad provider ordering while revealing differences in response quality and confidence behavior. Panel a reports final mean scores across the three judges for overall quality, clinical correctness, temporal grounding, evidence grounding, and calibration for the four fixed judge-phase contestants. Panel b shows pairwise cross-judge Spearman correlations for overall quality and clinical correctness. Panel c compares overall macro primary score with the fraction of high-confidence responses across all nine providers; all provider points are labeled. Confidence fraction uses full 53,070-prompt provider summaries where available and is the share of parsed responses with `confidence_value` equal to high, not statistical probability calibration. The judge packet contained 500 prompt instances and 2,000 contestant responses, evaluated by Claude Opus 4.6, GPT-5.4, and Gemini 3.1 Pro on the same rows.

Table 5 Judge-based provider summary for the four fixed contestants, reporting mean rubric scores across the three judges on a 1–5 scale.

Provider	Overall	Clinical	Temporal	Evidence	Calibration
GPT-5.4	4.397	4.383	4.722	4.655	4.325
DeepSeek Chat	3.894	3.882	4.342	4.167	3.819
Aloe-70B	2.823	2.885	3.464	3.209	3.008
MedGemma-4B	1.813	1.883	2.298	1.945	1.793

Table 6 Judge agreement summary across the shared 2,000-row judge packet.

Judge A	Judge B	Score	Rows	Spearman ρ	Exact match rate
Claude Opus 4.6	Gemini 3.1 Pro	overall quality	2000	0.783	0.283
Claude Opus 4.6	Gemini 3.1 Pro	clinical correctness	2000	0.759	0.286
Claude Opus 4.6	GPT-5.4	overall quality	2000	0.814	0.527
Claude Opus 4.6	GPT-5.4	clinical correctness	2000	0.808	0.511
Gemini 3.1 Pro	GPT-5.4	overall quality	2000	0.739	0.410
Gemini 3.1 Pro	GPT-5.4	clinical correctness	2000	0.719	0.411

4 Discussion

Four findings stand out. Temporal clinical reasoning difficulty was strongly condition-dependent (RQ2): delirium was near ceiling, whereas stroke remained difficult. Explicit temporal sequence modeling did not uniformly outperform a compact structured summary (RQ1). General-purpose models led the aggregate comparison, while medical-domain models remained below Tier 1A and Tier 1B groups overall (RQ3). Finally, trajectory-template support carried information about reasoning difficulty and should be reported in temporal clinical benchmarks (RQ2).

The branch comparison challenges the assumption that more temporal detail is always beneficial. Branch B1 improved AUROC on only 3 of 7 eligible tasks and AUPRC on only 1 of 7. The defensible interpretation is narrower: temporal structure helps when it aligns with the task’s decision boundary and is represented in a recoverable form. For other tasks, the anchor-state summary captured the relevant signal with less noise.

The aggregate tier gap is the most consequential benchmark-level result. All four Tier 2 medical-domain models underperformed the Tier 1A and Tier 1B groups in the aggregate macro comparison, although local condition-level rankings varied. This suggests that current medical adaptation is better aligned to medical language competence, knowledge recall, and static question answering than to anchor-bounded interpretation of evolving ICU evidence. The finding is consistent with broader clinical LLM evaluations showing that encoded medical knowledge does not automatically yield robust open-ended clinical decision making[36, 37], and with calls for clinically relevant validation of generalist medical AI systems[1, 19].

TIMELY Bench is complementary to existing benchmarks rather than a replacement for them. MedQA and PubMedQA test medical knowledge and biomedical reading comprehension[14, 15]; HealthBench, MedHELM, and LLMEval-Med broaden open-ended clinical evaluation[16–18]; EHRSHOT evaluates few-shot transfer on coded longitudinal records[13]; and MedAgentBench tests interactive tool use in a virtual EHR environment[20]. TIMELY Bench goes deeper on a narrower construct: interpreting synchronized ICU evidence under a masked information boundary. That emphasis explains why medical-domain models such as Me-LLaMA and MEDITRON, designed to strengthen clinical language competence and medical task performance[38, 39], did not necessarily outperform general-purpose models on this benchmark.

The comparison with workflow benchmarks is particularly important. An agent may retrieve the right chart fields and still reason incorrectly about their temporal order, while a model may answer a fixed prompt well but fail in interactive search. TIMELY Bench deliberately removes navigation as a source of variation so that errors can be attributed to temporal interpretation, evidence grounding, or task definition rather than to tool-use behavior.

This distinction matters for how the results should be used. TIMELY Bench should not be read as a general ranking of medical usefulness, nor as evidence that domain-specific training

is unnecessary. It shows that domain adaptation alone was insufficient for this particular composite skill: bounded longitudinal interpretation across structured data, notes, treatments, and condition-specific task definitions. Future domain adaptation may need to include longitudinal supervision, event-time objectives, and explicit evidence-grounding targets rather than only medical corpora or question-answering data.

CRES made the failure modes more interpretable. D3 trend reasoning was comparatively strong, D2 threshold reasoning separated tiers sharply, and D4 diagnostic discrimination was weak across the board. Current models can often detect directional change once the relevant signal is explicit, but struggle when the task requires selecting among overlapping clinical explanations. Confidence behavior also mattered. Gemini 3.1 Pro achieved the highest overall macro score but marked 99.2% of prompts as high confidence, compared with 69.9% for GPT-5.4. This is not statistical calibration, but it is relevant for deployment-facing evaluation because over-compressed confidence labels limit selective escalation and uncertainty-aware review.

The clinical significance of TIMELY Bench is not that it identifies a deployable model. It provides a pre-deployment evaluation scaffold for asking whether a model respects an ICU information boundary, recognizes threshold crossing, distinguishes atypical or low-support trajectories, and attributes its answer to visible evidence. This logic is relevant beyond the four conditions studied here. Trial recruitment, endpoint abstraction, pharmacovigilance, and rare disease natural-history work increasingly use longitudinal EHR data, and each depends on whether progression or safety events are identifiable from evidence available at the relevant time[40–45]. TIMELY Bench does not validate models for those operational uses, but it offers a testable structure for future disease-specific extensions.

For example, a trial eligibility screen should not infer an endpoint from information that became available after enrolment, and a safety-review assistant should distinguish a transient laboratory abnormality from a temporally evolving adverse event. The same principle applies to rare disease phenotyping, where evidence is sparse and progression may be distributed across notes, laboratory trends, and treatment changes. These are future extensions, not claims established by the present ICU benchmark, but they illustrate why anchor-bounded evaluation has practical value.

In that sense, the benchmark is closer to a diagnostic instrument than to a deployment scorecard. It can reveal whether a candidate model fails because it ignores the temporal boundary, cannot map measurements to thresholds, overuses confidence labels, or lacks sufficient evidence grounding. Those distinctions are clinically meaningful because the mitigation differs in each case: better serialization, different supervision, human review, or exclusion from a particular workflow.

Trajectory-template support is a methodological contribution in its own right. Delirium, sepsis, and AKI differed not only in performance, but in how often stays could be mapped onto executable physiology templates. The gap between delirium (96% stay-level support) and AKI (6%) indicates that disease-specific temporal regularity is part of the benchmark construct.

Future temporal clinical benchmarks should report both task distributions and how much of the trajectory space is clinically templatable.

Several limitations follow from the study design. TIMELY Bench is retrospective and derived from one source environment, so it does not establish transportability across institutions, documentation cultures, or EHR systems. Stroke bypasses the B3 template layer, so the four condition families are not perfectly symmetric at the latent-state level. Judge evaluation is model-based rather than human-adjudicated; LLM-as-judge methods remain a complement to, not a substitute for, expert review[17, 35]. Cross-condition causal pathways such as septic AKI and septic encephalopathy are not modeled. The delirium anchor relies on CAM-ICU assessments (item 228332), which may underrepresent hypoactive delirium. SOFA decomposition for sepsis currently covers four of six organ components; neurological and coagulation components should be added in future iterations. The benchmark also does not include prospective evaluation in live clinical workflows. Any future interventional use would require prospective protocols and reporting aligned with AI-specific guidance such as SPIRIT-AI and CONSORT-AI[46, 47].

Taken together, these results support TIMELY Bench as a diagnostic instrument for temporal clinical AI evaluation. It frames temporal reasoning as a condition-aware, anchor-bounded, multi-representation problem, exposes when temporal structure helps and when it does not, and makes disease-specific trajectory support visible. That role is narrower than clinical deployment, but broader than a leaderboard.

5 Conclusion

TIMELY Bench provides a reproducible framework for evaluating temporal clinical reasoning under realistic ICU information constraints. Across four condition families, explicit hourly sequence modeling helped only selectively, general-purpose LLMs led the aggregate provider comparison, and CRES analysis revealed failure modes hidden by overall scores. The benchmark should not be interpreted as validating any model for clinical deployment. Its value is as a pre-deployment diagnostic instrument: it tests whether models respect an anchor-bounded evidence state, identify temporal and threshold events, handle atypical or low-support trajectories, and ground answers in visible clinical evidence. Future work should test transportability across institutions, add multimorbidity and cross-condition causal pathways, and evaluate whether targeted longitudinal supervision improves these capabilities.

Data availability

The source clinical data used in this study are available through the PhysioNet credentialed data access program and BigQuery-hosted MIMIC-IV v3.1 and MIMIC-IV Note releases (MIMIC-IV, <https://physionet.org/content/mimiciv/>). Derived benchmark artifacts, including task manifests, evaluation samples, and scoring outputs, will be released upon publication.

Code availability

The benchmark assembly, prompt serialization, and scoring code will be released in a public repository upon publication.

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Author contributions

H.W., Z.L., L.Q., and Z.I. contributed equally to the conception and design of the study, benchmark development, analysis and interpretation of results, figure design, and manuscript drafting and revision. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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Appendix A. Illustrative TIMELY Bench prompt instance

Appendix A illustrates the Branch B2 prompt format using an AKI-T1 D1 instance at anchor hour 11. This is a negative example because no AKI Stage 2+ onset occurs within the 24-hour horizon after the anchor. GPT-5.4 correctly abstains from assigning an onset hour, whereas MedGemma-4B overcalls Hour 4 from transient hypotension.

Appendix Box A1 | Example prompt instance and model responses.

Prompt structure (AKI-T1, D1, anchor = Hour 11)

```
[SYSTEM]
Reason step by step, cite measurements and
timestamps, and answer based ONLY on the
data provided.
[PATIENT CONTEXT]
=== PATIENT CONTEXT (Condition: aki, Task:
AKI-T1, Anchor: Hour 11.0) ===
=== SECTION 1: DEMOGRAPHICS & COMORBIDITIES
===
62-year-old M. ckd.
=== SECTION 2: STRUCTURED TIMELINE ===
[Hour 0] map_merged=92.0; spo2=98.0;
sofa_total=0
[Hour 1] heart_rate=82.0; map_merged=79.0;
spo2=100.0; rrt_active=1; sofa_total=0
[Hour 4] heart_rate=73.0; map_merged=62.0;
spo2=100.0; rrt_active=1; sofa_total=1
[Hour 7] heart_rate=81.0; map_merged=121.0;
spo2=99.0; urineoutput_hourly=200.0;
rrt_active=1; sofa_total=1
... truncated; prompt continues through Hour
11 ...
=== SECTION 3: MEDICATIONS ===
No events in window.
=== SECTION 4: PROCEDURES ===
No events in window.
=== SECTION 5: CLINICAL NOTES ===
[Hour 0] [nursing] Patient Verbalized
... truncated; prompt contains additional
nursing lines through Hour 9 ...
[QUESTION]
At what hour did the earliest sign of AKI
Stage 2 progression appear?
[FORMAT]
{reasoning, answer, evidence[], confidence}
```

Strong response (GPT-5.4; correct)

```
"reasoning": "Stage 2 progression would
require creatinine worsening or sustained
oliguria. No creatinine values are visible,
urine output is preserved, and RRT is already
active from Hour 1; therefore the window
contains no new identifiable Stage 2 signal."
"answer": "No clear hour identified."
"evidence": [
  {"timestamp": "Hour
1", "measurement": "rrt_active", "value": "1"},
  {"timestamp": "Hour 7", "measurement": "urine
output", "value": "200 mL"}
]
"confidence": "low"
```

Weak response (MedGemma-4B; incorrect)

```
"reasoning": "MAP fell from 93 to 62 by Hour
4, suggesting worsening renal perfusion and
AKI progression."
"answer": "Hour 4"
"evidence": [{"timestamp": "Hour
4", "measurement": "MAP", "value": "62"}]
"confidence": "high"
```

The example uses a frozen AKI-T1 D1 prompt from the formal evaluation packet and contrasts a strong temporal localization response with a weak one on the same anchor-bounded evidence state. Patient-context evidence lines are copied from the frozen prompt, while system and question text are lightly abbreviated for readability. Full full_multimodal prompts averaged approximately 1,978 tokens in the frozen 12k evaluation sample.